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CSE - Shift - I (2nd Year - 2023)

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COMPUTATIONAL METHODS USING C++

(2023-24) Session — 3rd Sem

(Complete CM-Lab File)

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PROGRAM - 01.

Q → WAP To Implement Bisection Method in C++?

```
→ #include <iostream>
#include <stdlib.h>
#include <iomanip>
#include <cmath>
using namespace std;
```

// Function To Calculate Value For $(\cos(x) - x e^x)$ Values....

```
double f (double x) {
    double a =  $\cos(x) - x * \exp(x)$ ;
    return a;
```

```
}
```

```
int main () {
```

```
    int iter = 1; cout << setprecision (6) << fixed;
```

```
    double a, b, c, fa, fb, fc;
```

```
    double e = 0.00001;
```

```
    cout << "Enter The Start Value of Guess ::: ";
```

```
    cin >> a;
```

```
    cout << "Enter The End Value of Guess ::: ";
```

```
    cin >> b;
```

```
    fa = f(a);
```

```
    fb = f(b);
```

```
    if (fa * fb > 0.0) {
```

```
        cout << "You Entered The Incorrect Guesses " << endl;
```

```
    }
```

Teacher's Signature: _____

else {

do {

 $c = (a + b) / 2;$ $f_c = f(c);$ $\text{cout} << \text{" Iteration - " << iter << " It x = " << setw(10);$ $\text{cout} << c << \text{" And } f(c) = " << setw(10) << f_c << \text{endl};$ if ($f_a * f_c < 0$) { $b = c;$ }else { $a = c;$ } $iter++;$ } while ($f_a * f_c > e$); $\text{cout} << \text{" Root For Equation '(} \cos x - x e^x \text{)' is :: " << c << \text{endl};$

}

return 0;

{

CM1_BisectionMethod.cpp

```
1 //Code To Implement Bisection Method Using C++.....
2 #include<iostream>
3 #include<stdlib.h>
4 #include<iomanip>
5 #include<cmath>
6 using namespace std;
7
8 // double f(double x); //Function Prototype.....
9
10 //Defined the Function To Calculate Function Values For(cos(x)-x*exp(x)).....
11 double f(double x) {
12     double a=cos(x)-x*exp(x); //write the equation(cos(x)-x*exp(x)) whose roots are to be determined....
13     return a;
14 }
15
16 int main() {
17     int iter=1;
18     double a,b,c,e,fa,fb,fc; //declare some needed variables.....
19     cout<<setprecision(6)<<fixed; //Set The Precision Of Our Output...
20     e=0.00001;
21     cout<<"ENTER THE START VALUE OF GUESS ::: ";
22     cin>>a;
23     cout<<"ENTER THE END VALUE OF GUESS ::: ";
24     cin>>b;
25     cout<<endl;
26     fa=f(a),fb=f(b);
27     if(fa*fb>0.0){
28         cout<<"YOU ENTERED INCORRECT GUESSES"<<endl;
29     }
30     else{
31         do{
32             c=(a+b)/2;
33             fc=f(c);
34             cout<<"Iteration-"<<iter<<"\tx = "<<setw(10)<<c<<" And f(x) = "<<setw(10)<<fc<<endl;
35             if(fa*fc<0){ b=c; }
36             else{ a=c; }
37             iter++;
38         }while (fabs(fc)>e);
39         cout<<endl<<"Root For The Equation '(cos(x) - x*exp(x))' IS ::: "<<c<<endl;
40     }
41     return 0;
42 }
43
44 //For Theory --> https://youtu.be/3j0c\_Fh0t5U?si=UzII0qoyqbvocvJG
45 //For Code ---> https://youtu.be/H1i0o040u3c?si=S7LuXOLW8-N2n6Vs
```

Snipped

OUTPUT :-

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ISHA PANDEY\YashPandey_Codes(DSA)> cd "c:\Users\ISHA PANDEY\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes\" ;  
CM1_BisectionMethod }
```

ENTER THE START VALUE OF GUESS ::: 0

ENTER THE END VALUE OF GUESS ::: 1

Iteration-1	x =	0.500000	And	f(x) =	0.053222
Iteration-2	x =	0.750000	And	f(x) =	-0.856061
Iteration-3	x =	0.625000	And	f(x) =	-0.356691
Iteration-4	x =	0.562500	And	f(x) =	-0.141294
Iteration-5	x =	0.531250	And	f(x) =	-0.041512
Iteration-6	x =	0.515625	And	f(x) =	0.006475
Iteration-7	x =	0.523438	And	f(x) =	-0.017362
Iteration-8	x =	0.519531	And	f(x) =	-0.005404
Iteration-9	x =	0.517578	And	f(x) =	0.000545
Iteration-10	x =	0.518555	And	f(x) =	-0.002427
Iteration-11	x =	0.518066	And	f(x) =	-0.000940
Iteration-12	x =	0.517822	And	f(x) =	-0.000197
Iteration-13	x =	0.517700	And	f(x) =	0.000174
Iteration-14	x =	0.517761	And	f(x) =	-0.000012
Iteration-15	x =	0.517731	And	f(x) =	0.000081
Iteration-16	x =	0.517746	And	f(x) =	0.000035
Iteration-17	x =	0.517754	And	f(x) =	0.000011
Iteration-18	x =	0.517757	And	f(x) =	-0.000000

Root For The Equation '(cos(x) - x*exp(x))' IS ::: 0.517757

```
PS C:\Users\ISHA PANDEY\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes> |
```

PROGRAM - 02.

Q → WAP To Implement Newton Raphson Method using C++?

→ #include <iostream>

#include <bits/stdc++.h>

using namespace std;

double func (double x) { return (x*x*x) - (4*x) - 9; }

double funcder (double x) { return ((3*x*x) - 4); }

void NewtonRaphson (double x1) { // Fⁿ To Implement Method.....

double h = func (x1) / funcder (x1);

while (fabs(h) >= 0.0001) {

h = func (x1) / funcder (x1);

x1 = x1 - h;

}

cout << "The Value of Root is ::: " << x1 << endl;

}

int main () {

double x0 = -20;

NewtonRaphson (x0); // Calling The Fⁿ.....

return 0;

}

∴ OUTPUT → The Value of Root is ::: 2.70653

Teacher's Signature: _____

CM2_NewtonRhapson[1].cpp

```
1  #include <iostream>
2  #include <bits/stdc++.h>
3  using namespace std;
4
5  double func(double x){
6      return x*x*x - 4*x - 9;
7  }
8
9  double funcDerv(double x){
10     return 3*x*x - 4;
11 }
12
13 void NewtonRhapson(double x1){
14     double h = func(x1)/funcDerv(x1);
15     int itr=1;
16     while( abs(h) >= 0.0001 ){
17         h = func(x1)/funcDerv(x1);
18         x1 = x1 - h;
19         cout<<"Iteration-"<<itr<<" , The Value Of Root Is ::: "<<x1<<endl;
20         itr++;
21     }
22     cout << "THE VALUE OF ROOT IS ::: " << x1;
23 }
24
25 int main() {
26     cout<<endl;
27     double Xo = -20;
28     NewtonRhapson(Xo);
29     cout<<endl;
30     return 0;
31 }
32
```

Snipped

OUTPUT :-

```
PROBLEMS 104 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes\" ;
on }

Iteration-1 , The Value Of Root Is ::: -13.3704
Iteration-2 , The Value Of Root Is ::: -8.96367
Iteration-3 , The Value Of Root Is ::: -6.03865
Iteration-4 , The Value Of Root Is ::: -4.09316
Iteration-5 , The Value Of Root Is ::: -2.77017
Iteration-6 , The Value Of Root Is ::: -1.76199
Iteration-7 , The Value Of Root Is ::: -0.365194
Iteration-8 , The Value Of Root Is ::: -2.47301
Iteration-9 , The Value Of Root Is ::: -1.48103
Iteration-10 , The Value Of Root Is ::: 0.970006
Iteration-11 , The Value Of Root Is ::: -9.19538
Iteration-12 , The Value Of Root Is ::: -6.19242
Iteration-13 , The Value Of Root Is ::: -4.19594
Iteration-14 , The Value Of Root Is ::: -2.84214
Iteration-15 , The Value Of Root Is ::: -1.82453
Iteration-16 , The Value Of Root Is ::: -0.525732
Iteration-17 , The Value Of Root Is ::: -2.74673
Iteration-18 , The Value Of Root Is ::: -1.74124
Iteration-19 , The Value Of Root Is ::: -0.305863
Iteration-20 , The Value Of Root Is ::: -2.4044
Iteration-21 , The Value Of Root Is ::: -1.40896
Iteration-22 , The Value Of Root Is ::: 1.74179
Iteration-23 , The Value Of Root Is ::: 3.83585
Iteration-24 , The Value Of Root Is ::: 3.03627
Iteration-25 , The Value Of Root Is ::: 2.74688
Iteration-26 , The Value Of Root Is ::: 2.70724
Iteration-27 , The Value Of Root Is ::: 2.70653
Iteration-28 , The Value Of Root Is ::: 2.70653
THE VALUE OF ROOT IS ::: 2.70653
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes>
```

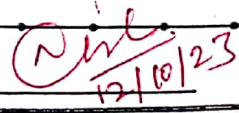
Program - 3.

0 → WAP To Implement Secant Method Using C++ ?

```

→ #include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;
double f(double x) {
    return pow(x, 3) - (5 * x) + 1.0; //  $x^3 - 5x + 1$ 
}
int main() {
    cout << precision(4);
    cout << setf(ios::fixed); int it = 1;
    double a, b, c, fa, fb, fc;
    cout << "Enter The Initial N.O. of Guesses :: "; cin >> b;
    cout << "Enter The Initial N.O. of Guesses :: "; cin >> c;
    cout << "Enter The Degree of Accuracy :: "; cin >> e;
    do {
        a = b; b = c; fa = f(a); fb = f(b);
        c = ((a * fb) - (b * fa)) / (fb - fa);
        cout << "The Value of c = " << c << " & f(c) = " << f(c) << endl;
        it++;
        if (f(c) == 0) {
            cout << "The Value of c = " << c << " & f(c) = " << f(c) << endl;
            return 0;
        }
    } while (fabs(c - b) >= e);
    cout << "The Root of  $x^3 - 5x + 1$  is :: " << c << " & f(c) = " << f(c) << endl;
    return 0;
}

```

Teacher's Signature: 

CM3_SecantMethod[1].cpp

```
1 //Secant Method(Chord Method) for finding the roots of an equation...
2 #include<iostream>
3 #include<iomanip>
4 #include<cmath>
5 using namespace std;
6
7 double f(double x); //declare the function for the given equation.....
8
9 double f(double x) { //define the function here, ie give the equation.....
10     double a=pow(x,3)-(5*x)+1.0; //write the equation whose roots are to be determined.....
11     return a;
12 }
13
14 int main() {
15     cout.precision(4);
16     cout.setf(ios::fixed); //set the precision of the output.....
17
18     int itr=1;
19     double a,b,c,fa,fb,e;
20     cout<<"ENTER THE INITIAL No OF GUESSES ::: ";
21     cin>>b; //take an intial guess.....
22     cout<<"ENTER THE INITIAL No OF GUESSES ::: ";
23     cin>>c; //take an intial guess.....
24     cout<<"ENTER THE DEGREE OF ACCURACY ::: ";
25     cin>>e; //take the desired accuracy.....
26
27     do{
28         a=b; b=c; //make b equal to the last calculated value of c....
29         fa=f(a) ; fb=f(b);
30         c = ( (a*fb) - (b*fa) ) / (fb - fa); //calculate c.....
31         // c=b-(b-a) / ( (f(b)-f(a)) * f(b) ); //calculate c.....
32         cout<<"IN ITERATION-"<<itr<<" , THE VALUE OF c = "<<<<" & f(c) = "<<<f(c)<<endl;
33         itr++;
34
35         if (f(c)==0){
36             cout<<endl;
37             cout<<"THE ROOT OF THE EQUATION '(x^3) - (5x) + 1' IS ::: "<<<<" & f("<<<<") = "<<<f(c)<<endl; //print the root.....
38             return 0;
39         }
40
41     }while(fabs(c-b)>=e); //check if the error is greater than the desired accuracy.....
42
43     cout<<endl;
44     cout<<"THE ROOT OF THE EQUATION '(x^3) - (5x) + 1' IS ::: "<<<<" & f("<<<<") = "<<<f(c)<<endl; //print the root.....
45     return 0;
46 }
47
48 // Theory --> https://youtu.be/Eud16189QRA?si=20JX79hFT3wf111A
49 // Code --> https://youtu.be/TRFtrYdugdY?si=iRg5Rm8lwwKaGK4m
```

Snipped

OUTPUT : -

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes\" ;
}
ENTER THE INITIAL No OF GUESSES ::: 0
ENTER THE INITIAL No OF GUESSES ::: 1
ENTER THE DEGREE OF ACCURACY ::: 0.00001
IN ITERATION-1 , THE VALUE OF c = 0.2500 & f(c) = -0.2344
IN ITERATION-2 , THE VALUE OF c = 0.1864 & f(c) = 0.0743
IN ITERATION-3 , THE VALUE OF c = 0.2017 & f(c) = -0.0005
IN ITERATION-4 , THE VALUE OF c = 0.2016 & f(c) = -0.0000
IN ITERATION-5 , THE VALUE OF c = 0.2016 & f(c) = 0.0000

THE ROOT OF THE EQUATION '(x^3) - (5x) + 1' IS ::: 0.2016 & f(0.2016) = 0.0000
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes> █
```

SYLLABUS

Program - 4.

Q → WAP To Implement Lagrange's Interpolation method using C++?

```
→ #include <iostream>
#include <iomanip>
#include <cmath>
using namespace std;
```

```
class Data {
```

```
public: int x, y; // y = f(x).....
```

```
};
```

```
double Interpolate (Data f[], int xi, int n) {
```

```
double result = 0;
```

```
for (int i=0; i < n; i++) {
```

```
double term = f[i].y;
```

```
for (int j=0; j < n; j++) {
```

```
if (j != i) {
```

```
term = term * ((xi - f[j].x) / (f[i].x - f[j].x));
```

```
}
```

```
}
```

```
result += term;
```

```
}
```

```
return result;
```

```
}
```

```
int main () {
```

```
Data f[] = { {1,2}, {2,4}, {3,8}, {4,16}, {7,128} };
```

```
cout << " Value of f(5) is :: " << Interpolate (f,5,5) << endl;
```

```
return 0;
```

```
}
```

Teacher's Signature: _____


```

1 // C++ program for Implementation of Lagrange's Interpolation.....
2 #include<iostream>
3 #include<iomanip>
4 #include<cmath>
5 using namespace std;
6
7 // Class To represent a data point corresponding to 'x' and y = f(x).....
8 class Data{
9     public:
10         int x, y;
11 };
12
13 // Function to Interpolate the given data points using Lagrange's formula
14 // 'xi' which corresponds to the new data point whose value is to be obtained
15 // 'n' represents the number of known data points.....
16
17 double Interpolate(Data f[], int xi, int n){
18     double result = 0; // Initialize result.....
19     for (int i = 0 ; i < n ; i++) {
20         double term = f[i].y; // To Compute individual terms of above formula.....
21         for (int j = 0 ; j < n ; j++){
22             if (j != i){
23                 term = term * (xi - f[j].x) / double(f[i].x - f[j].x); //Type Casting....
24             }
25         }
26         result += term; // Add current term to result
27     }
28     return result;
29 }
30
31 int main(){
32     cout<<endl;
33     Data f[] = {{1,2}, {2,4}, {3,8}, {4,16}, {7,128}}; //Array Of Objects.....
34     // Using the Interpolate function to obtain a data point corresponding to x=5.....
35     cout << "Value of f(5) is : " << Interpolate(f, 5, 5);
36     cout<<endl;
37     cout<<endl;
38     return 0;
39 }
40
41 // Theory --> https://youtu.be/2h0R0Uka1HI?si=CzwXPHCtjrV8n97z
42

```

OUTPUT : -

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes\" ;  
; if ($?) { .\CM4_Lagrange_InterpolationMethod }
```

Value of $f(5)$ is : 32.9333

```
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes>
```

SYLLABUS

Program - 5.

Q → WAP To Implement Newton Divided Difference Interpolating Polynomial Method Using C++ language ?

→ #include <iostream>
using namespace std;

```
int main () {
    float x[100], y[100], xp, yp=0, b;
    int i, j, n;
    cout << "Enter The Number of Data ::: "; cin >> n;
    cout << "Enter The Values of Data for Calculation" << endl;
    for (int i = 1; i <= n; i++) {
        cout << "x[" << i << "] = "; cin >> x[i];
        cout << "y[" << i << "] = "; cin >> y[i];
    }
    cout << "Enter The Interpolating Points ::: "; cin >> xp;
    for (int i = 1; i <= n; i++) { // Nested loop for Calculation.....
        b = 1;
        for (int j = 1; j <= n; j++) {
            if (i != j) {
                b = b * (xp - x[j]) / (x[i] - x[j]);
            }
        }
        yp = yp + (b * y[i]);
    }
    cout << "Interpolated Value At " << xp << " Is ::: " << yp << endl;
    return 0;
}
```

Teacher's Signature: _____


```

1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      cout<<endl;
6      cout<<endl;
7      float x[100], y[100], xp, yp=0, p;
8      int i,j,n;
9
10     // Input Section For Input Of Variables.....
11     cout<<"ENTER THE NUMBER OF DATA ::: ";
12     cin>>n;
13     cout<<"ENTER THE VALUES OF DATA FOR CALCULATION"<< endl;
14     for(i = 1 ; i <= n ; i++) {
15         cout<<"x["<<i<<" ] = ";
16         cin>>x[i];
17         cout<<"y["<<i<<" ] = ";
18         cin>>y[i];
19     }
20     cout<<"ENTER THE INTERPOLATING POINTS ::: ";
21     cin>>xp;
22     /* Implementing Lagrange Interpolation */
23     for(i = 1 ; i <= n ; i++) {
24         p=1;
25         for(j=1;j<=n;j++) {
26             if(i!=j){
27                 p = p* (xp - x[j])/(x[i] - x[j]);
28             }
29         }
30         yp = yp + p * y[i];
31     }
32     cout<<endl;
33     cout<<"INTERPOLATED VALUE AT "<<xp<<" IS ::: "<<yp<<endl;
34     return 0;
35 }
36
37

```

OUTPUT :-



```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes\" ;
\newton_Divided_Difference }

ENTER THE NUMBER OF DATA ::: 6
ENTER THE VALUES OF DATA FOR CALCULATION
x[1] = 0
y[1] = 1
x[2] = 1
y[2] = 14
x[3] = 2
y[3] = 15
x[4] = 4
y[4] = 5
x[5] = 5
y[5] = 6
x[6] = 6
y[6] = 19
ENTER THE INTERPOLATING POINTS ::: 3

INTERPOLATED VALUE AT 3 IS ::: 10
PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes> |
```

Program - 6.

Q → WAP To Implement Trapezoidal Method in C++ language?

→ #include <iostream>
#include <iomanip>
#include <math.h>
using namespace std;

#define f(x) (1/(1+pow(x,2)))

int main() {

float lower, upper, integration = 0, stepsize, k;

int i, subInterval;

cout << "Enter The lower limit of Integration :: ";

cin >> lower;

cout << "Enter The upper limit of Integration :: ";

cin >> upper;

cout << "Enter The NO. of Sub-Intervals :: ";

cin >> subInterval;

stepsize = (upper - lower) / subInterval;

cout << "Value of h → " << setw(7) << stepsize << endl;

cout << setw(6) << "x" << setw(17) << "f(x)" << endl;

integration = f(lower) + f(upper);

cout << setw(10) << lower << setw(15) << upper << endl;

"ON NEXT PAGE"

Teacher's Signature: _____


```
for ( i = 1 ; i <= subInterval - 1 ; i++ ) {  
    k = lower + ( i * stepsize ) ;  
    cout << setw(10) << k << setw(15) << f(k) << endl ;  
    integration += ( 2 * f(k) ) ;  
}  
cout << setw(10) << upper << setw(15) << f(upper) << endl ;  
integration *= ( stepsize / 2 ) ;  
cout << "Required Value of Integration is :::" << integration << endl ;  
cout << endl ;  
return 0 ;
```

}

Teacher's Signature: _____

```

1  #include <iostream>
2  #include <math.h>
3  #include <iomanip>
4  using namespace std;
5
6  #define f(x) (1/(1+pow(x,2)))
7
8  int main(){
9      cout << endl;
10     float lower, upper, integration = 0.0, stepSize, k;
11     int i, subInterval;
12
13     /* Input */
14     cout<< "Enter lower limit of integration -> ";
15     cin >> lower;
16     cout << "Enter upper limit of integration -> ";
17     cin >> upper;
18     cout << "Enter number of sub intervals    -> ";
19     cin >> subInterval;
20
21     /* Finding step size */
22     stepSize = (upper - lower) / subInterval;
23     cout << endl;
24     cout<< "value of h -> " << setprecision(7) << stepSize << endl; // Set precision to 7 decimal place
25     s  cout<< endl;
26
27     // Print column headers
28     cout << setw(6) << "x" << setw(17) << "f(x)" << endl;
29     cout << fixed << setprecision(7); // Set precision to 7 decimal places
30
31     /* Finding Integration Value */
32     integration = f(lower) + f(upper);
33     cout << setw(10) << lower << setw(15) << f(lower) << endl;
34     for (i = 1; i <= subInterval - 1; i++) {
35         k = lower + i * stepSize;
36         cout << setw(10) << k << setw(15) << f(k) << endl;
37         integration = integration + 2 * (f(k));
38     }
39     cout << setw(10) << upper << setw(15) << f(upper) << endl;
40     integration = integration * stepSize / 2;
41     cout<< "Required value of integration is: " << integration << endl;
42     cout << endl;
43     return 0;
44 }
45

```

OUTPUT :-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes\" ;
rapezoidal_Method }

Enter lower limit of integration -> 0
Enter upper limit of integration -> 1
Enter number of sub intervals    -> 12

value of h -> 0.08333334

      x          f(x)
0.0000000    1.0000000
0.0833333    0.9931034
0.1666667    0.9729730
0.2500000    0.9411765
0.3333333    0.9000000
0.4166667    0.8520710
0.5000000    0.8000000
0.5833334    0.7461140
0.6666667    0.6923077
0.7500000    0.6400000
0.8333334    0.5901639
0.9166667    0.5433962
1.0000000    0.5000000
Required value of integration is: 0.7851088

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes> |
```


Program - 7.

Q → WAP To Implement Simpson - $(1/3)$ - Rule in C++ ?

→ #include <iostream>
#include <cmath>
#include <iomanip>
using namespace std;

#define f(x) (1/(1+x)) // Function Expression
#define I(x) (log(1+x)) // Integrated Expression
float Y[5] = {0.0};

```
void Display (float h, float start) {  
    cout << "x" << "\t" << setprecision(10) << " f(x)" << endl;  
    float Num = start;  
    for (int i=0; i<=4; i++) {  
        cout << setprecision(5) << Num << "\t" << setprecision(10)  
            << Y[i] << endl;  
        Num += h;  
    }  
    cout << endl;  
}
```

```
int main () {  
    float a, b, h;  
    cout << "Enter The start value of Integration :::" << endl;  
    cin >> a;  
    cout << "Enter The End value of Integration :::" << endl;  
    cin >> b;  
    h = (b-a)/4;
```

Teacher's Signature: _____

```
for (int i=0 ; i<=4 ; i++) {  
    Y[i] = f(temp); temp += h;  
}  
Display (h, a);  
float Vol1 = (h/3)* ( (Y[0] + Y[4]) + 4*(Y[1] + Y[3])  
                    + 2*(Y[2]) );  
float Vol2 = I(b) - I(a);  
cout << " Approx Value of Integration :: " << Vol1 << endl;  
cout << " Actual Value of Integration :: " << Vol2 << endl;  
float Error = (Vol1 > Vol2) ? (Vol1 - Vol2) : (Vol2 - Vol1);  
cout << " The Error in The Approx Value Is :: " << Error << endl;  
return 0;  
}
```

Teacher's Signature: _____


```

1 //Code To Implement Simpson-(1/3)-Rule Using C++.....
2 #include<iostream>
3 #include <math.h>
4 #include <iomanip>
5 using namespace std;
6
7 #define f(x) (1/(1+x)) // Function Expression.....
8 #define I(x) (log(1+x)) // Integrated Function.....
9 float Y[5];
10
11 void Display(float h,float Start){
12     cout<<"x"<<"\t" <<setprecision(10)<<"f(x)"<<endl;
13     float Num=Start;
14     for(int i=0;i<=4;i++){
15         cout<<setprecision(5)<<Num<<"\t"<<setprecision(10)<<Y[i]<<endl;
16         Num =Num+h;
17     }
18     cout<<endl;
19 }
20
21 int main(){
22     cout<<endl;
23     float a,b,h;
24
25     // Statements To Input The Range From The User.....
26     cout<<"ENTER THE START VALUE FOR INTEGRATION ::: ";
27     cin>>a;
28     cout<<"ENTER THE END VALUE FOR INTEGRATION ::: ";
29     cin>>b;
30     cout<<endl;
31     h=(b-a)/4;
32
33     float temp=a;
34     for(int i=0;i<=4;i++){
35         Y[i]=f(temp); temp += h;
36     }
37
38     Display(h,a);
39
40     // Approx Value Calculated By Simpson-(1/3)-Rule.....
41     float Approx_f_x=(h/3)*((Y[0]+Y[4]) + 4*(Y[1]+Y[3]) + 2*(Y[2]));
42
43     // Actual Value Calculated By Integration.....
44     float Actual_f_x=I(b)-I(a);
45
46     cout<<"APPROX VALUE OF INTEGRATION IS ::: "<<setw(8)<<Approx_f_x<<endl;
47     cout<<"ACTUAL VALUE OF INTEGRATION IS ::: "<<setw(8)<<Actual_f_x<<endl;
48
49     float Error= (Actual_f_x>Approx_f_x) ? (Actual_f_x-Approx_f_x) : (Approx_f_x-Actual_f_
50 x); cout<<"THE ERROR IN THE APPROX VALUE IS ::: |Actual-Approx| i.e "<<Error<<endl;
51     cout<<endl;
52     return 0;
53 }

```

OUTPUT: -

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes\" ;
?) { .\CM09_Simpson_OneThird_Method }

ENTER THE START VALUE FOR INTEGRATION ::: 0
ENTER THE END VALUE FOR INTEGRATION ::: 1

x      f(x)
0      1
0.25   0.8000000119
0.5    0.6666666865
0.75   0.571428597
1      0.5

APPROX VALUE OF INTEGRATION IS ::: 0.693253994
ACTUAL VALUE OF INTEGRATION IS ::: 0.6931471825
THE ERROR IN THE APPROX VALUE IS ::: |Actual-Approx| i.e 0.0001068115234

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes> |
```


Program - 08.

Q → WAP To Implement Simpson - (3/8) - Rule Using C++?

```
→ #include <iostream>
#include <cmath>
#include <iomanip>
using namespace std;
```

```
#define f(x) (2/(1+pow(x,2))) // Function Expression.....
```

```
#define I(x) atan(x) // Integrated Function.....
```

```
float Y[7] = {0.0};
```

```
void Display (float h, float start) {
```

```
    cout << "x" << "t" << setprecision(10) << "f(x)" << endl;
```

```
    float Num = start;
```

```
    for (int i=0; i<=6; i++) {
```

```
        cout << setprecision(5) << Num << "t" << setprecision(10) << Y[i] << endl;
```

```
        Num += h;
```

```
    }
```

```
    cout << endl;
```

```
}
```

```
int main () {
```

```
    float a, b, h;
```

```
    cout << "Enter The start value of Integration ::: ";
```

```
    cin >> a;
```

```
    cout << "Enter The End value of Integration ::: ";
```

```
    cin >> b;
```

```
    h = (b-a)/6;
```

Teacher's Signature: _____

```
float temp = a;
for (int i=0 ; i <= 6 ; i++) {
    Y[i] = f(temp) ; temp += h;
}
Display ( h, a);
float Vol1 = (3*h/8) * ( (Y[0] + Y[6]) + 2*(Y[3]) +
                        3*(Y[1] + Y[2] + Y[4] + Y[5]) );
float Vol2 = I(b) - I(a);
cout << " Approx Value of Integration Is :::" << Vol1 << endl;
cout << " Actual Value of Integration Is :::" << Vol2 << endl;
float Error = (Vol1 > Vol2) ? (Vol1 - Vol2) : (Vol2 - Vol1);
cout << " The Error in The Approx Value Is :::" << Error << endl;
return 0;
```

}

Teacher's Signature: _____


```

1 //Code To Implement Simpson-(3/8)-Rule Using C++.....
2 #include<iostream>
3 #include <math.h>
4 #include <iomanip>
5 using namespace std;
6
7 #define f(x) (1/(1+pow(x,2))) // Function Expression.....
8 #define I(x) atan(x) // Integrated Function.....
9 float Y[7];
10
11 void Display(float h,float Start){
12     cout<<"x"<<"\t" <<setprecision(10)<<"f(x)"<<endl;
13     float Num=Start;
14     for(int i=0;i<=6;i++){
15         cout<<setprecision(5)<<Num<<"\t" <<setprecision(10)<<Y[i]<<endl;
16         Num =Num+h;
17     }
18     cout<<endl;
19 }
20
21 int main(){
22     cout<<endl;
23     float a,b,h;
24
25     // Statements To Input The Range From The User.....
26     cout<<"ENTER THE START VALUE FOR INTEGRATION ::: ";
27     cin>>a;
28     cout<<"ENTER THE END VALUE FOR INTEGRATION ::: ";
29     cin>>b;
30     cout<<endl;
31     h=(b-a)/6;
32
33     float temp=a;
34     for(int i=0;i<=6;i++){
35         Y[i]=f(temp); temp += h;
36     }
37
38     Display(h,a);
39
40     // Approx Value Calculated By Simpson-(3/8)-Rule.....
41     float Approx_f_x=(3*h/8)*((Y[0]+Y[6]) + 3*(Y[1]+Y[2]+Y[4]+Y[5]) + 2*(Y[3]));
42
43     // Actual Value Calculated By Integration.....
44     float Actual_f_x=I(b)-I(a);
45
46     cout<<"APPROX VALUE OF INTEGRATION IS ::: " <<setw(8)<<Approx_f_x<<endl;
47     cout<<"ACTUAL VALUE OF INTEGRATION IS ::: " <<setw(8)<<Actual_f_x<<endl;
48
49     float Error= (Actual_f_x>Approx_f_x) ? (Actual_f_x-Approx_f_x) : (Approx_f_x-Actual_f_
50 x); cout<<"THE ERROR IN THE APPROX VALUE IS ::: |Actual-Approx| i.e " <<Error<<endl;
51     cout<<endl;
52     return 0;
53 }

```


OUTPUT: -

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes\" ;
f ($?) { .\CM10_Simpson_ThreeEight_Method }

ENTER THE START VALUE FOR INTEGRATION ::: 0
ENTER THE END VALUE FOR INTEGRATION ::: 6

x      f(x)
0      1
1      0.5
2      0.200000003
3      0.1000000015
4      0.05882352963
5      0.03846153989
6      0.02702702768

APPROX VALUE OF INTEGRATION IS ::: 1.357080817
ACTUAL VALUE OF INTEGRATION IS ::: 1.405647635
THE ERROR IN THE APPROX VALUE IS ::: |Actual-Approx| i.e 0.04856681824

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes> |
```

PROGRAM - 09.

Q → WAP To find Inverse of System of Linear System Using Gauss Jordan Method?

→ #include <iostream>
using namespace std;

```
void PrintMatrix (double Matrix [3][6], int row, int col) {
    for (int i=0 ; i < row ; i++) {
        for (int j=0 ; j < col ; j++) {
            cout << Matrix [i][j] << endl;
        }
        cout << endl;
    }
}
```

```
void SetDiagonal (double Matrix [3][6], int row, int pivot) {
    for (int i=0 ; i < 6 ; i++) {
        Matrix [row][i] = Matrix [row][i] / pivot;
    }
}
```

```
void SetElement (double Matrix [3][6], int Currrow, int linetrow, double pivot, double curr) {
    double temp = curr / pivot;
    for (int i=0 ; i < 6 ; i++) {
        if (Matrix [Currrow][i] == curr) { Matrix [Currrow][i] = 0; }
        else {
            Matrix [Currrow][i] -= ((temp) * (Matrix [linetrow][i]));
        }
    }
}
```

Teacher's Signature: _____

```
void GaussJordan (double Matrix [3][6]) {  
    for (int Col=0; Col < 3; Col++) {  
        for (int Row=0; Row < 3; Row++) {  
            if (Row == Col) {  
                if (Matrix [Row] [Col] != 1.0) {  
                    SetDiagonal (Matrix, Row, Col, Matrix [Col] [Col],  
                                , Matrix [Row] [Col]);  
                }  
            }  
            else {  
                if (Matrix [Row] [Col] != 0.0) {  
                    SetElement (Matrix, Row, Col, Matrix [Col] [Col],  
                                , Matrix [Row] [Col]);  
                }  
            }  
        }  
    }  
}  
  
int main () {  
    cout << endl;  
    double Matrix [3][6] = { {1, 2, -2, 1, 0, 0}, {-1, 3, 0, 0, 1, 0}, {0, -2, 1, 0, 0, 1} };  
    cout << "The Original Matrix is :::" << endl;  
    PrintMatrix (Matrix, 3, 6);  
    cout << endl;  
    GaussJordan (Matrix);  
    cout << endl;  
    cout << "The Inverse of Matrix is :::" << endl;  
    PrintMatrix (Matrix, 3, 6);  
    cout << endl;  
    return 0;  
}
```

Teacher's Signature: _____


```

1 // WAP to Find Inverse of a matrix using Gauss Jordan method in C++ language.....
2 #include <iostream>
3 using namespace std;
4
5 void PrintMatrix(double Matrix[3][6], int rows, int cols) {
6     for (int i = 0; i < rows; i++) {
7         for (int j = 0; j < cols; j++) {
8             cout<<Matrix[i][j]<<"\t\t";
9         }
10        cout << endl;
11    }
12 }
13
14 void SetDiagonal(double Matrix[3][6],int row,double Pivot){
15     for(int i=0;i<6;i++){
16         Matrix[row][i] = Matrix[row][i]/Pivot;
17     }
18 }
19
20 void SetElement(double Matrix[3][6],int CurrRow,int PivotRow,double Pivot,double curr){
21     double temp=curr/Pivot;
22     for(int i=0;i<6;i++){
23         if(Matrix[CurrRow][i]==curr){ Matrix[CurrRow][i]=0; }
24         else{ Matrix[CurrRow][i] -= ((temp)*(Matrix[PivotRow][i])); }
25     }
26 }
27
28 void GaussJordan(double Matrix[3][6]) {
29     for(int Col=0;Col<3;Col++){
30         for(int Row=0;Row<3;Row++){
31             if(Row == Col){
32                 if(Matrix[Row][Col]!=1.0){ SetDiagonal(Matrix,Row,Matrix[Row][Col]); }
33             }
34             else{
35                 if(Matrix[Row][Col]!=0.0){ SetElement(Matrix,Row,Col,Matrix[Col][Col],Matrix[Row][Col]); }
36             }
37         }
38     }
39 }
40
41 int main() {
42     double Matrix[3][6] = {{1,2,-2,1,0,0},{-1,3,0,0,1,0},{0,-2,1,0,0,1}};
43     cout << "THE ORIGINAL MATRIX IS ::: " << endl;
44     PrintMatrix(Matrix, 3, 6);
45     GaussJordan(Matrix);
46     cout<<endl;
47     cout << "THE INVERSE MATRIX IS ::: " << endl;
48     PrintMatrix(Matrix, 3, 6);
49     return 0;
50 }

```

OUTPUT: -

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH TERMINAL OUTPUT  COMMENTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes\" ;
GaussJordanMethod }

THE ORIGINAL MATRIX IS :::
1      2      -2      1      0      0
-1     3      0      0      1      0
0     -2      1      0      0      1

THE INVERSE MATRIX IS :::
1      0      0      3      2      6
0      1      0      1      1      2
0      0      1      2      2      5

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes>
```

PROGRAM - 10.

Q → WAP To find Eigen Values using Power method?

→ #include <iostream>
#include <cmath>
using namespace std;

```
double GetMax (double Temp [3][1]) {  
    double Max = -1;  
    for (int i=0; i<3; i++) {  
        if (Temp [i] > Max) { Max = Temp [i]; }  
    }  
    return Max;  
}
```

```
void Multiply (double A [3][3], double X [3][1], double Result [3][1]) {  
    for (int i=0; i<3; i++) { Result [i][0] = 0; }  
    for (int i=0; i<3; i++) {  
        for (int j=0; j<3; j++) {  
            Result [i][0] += ((A[i][j]) * (X[j][0]));  
        }  
    }  
}
```

```
void SetVector (double Vector [3][1], double lambda) {  
    for (int i=0; i<3; i++) {  
        Vector [i][0] = (Vector [i][0] / lambda);  
    }  
}
```

Teacher's Signature: _____


```

void Update (double X [3] [1], double Result [3] [1]) {
    for (int i=0 ; i<3 ; i++) {
        X [i] [0] = Result [i] [0];
    }
}

```

```

void Display (int it, double NewLamda, double X [3] [1]) {
    cout << " It - " << it << "\t\t" << NewLamda << "\t\t";
    for (int i=0 ; i<3 ; i++) {
        if (i==2) { cout << X [i] [0] << "]" << endl; }
        else { cout << X [i] [0] << ", " ; }
    }
}

```

```

int main () {
    return 0; cout << endl;
    double A [3] [3] = { {2, -1, 0}, {-1, 2, -1}, {0, -1, 2} };
    double X [3] [1] = { {1}, {0}, {0} };
    double Vector [3] [1] = { 0 };
    double OldLamda = 1.0 , NewLamda = 0.0 ;
    double e = 0.001, Diff = (NewLamda - OldLamda);
    cout << " It " << "\t\t Eigen Value \t\t Eigen Vector " << endl;
    cout << endl;
    while (Diff >= e) {
        Multiply (A, X, Vector); OldLamda = NewLamda;
        NewLamda = GetMax (Vector);
        SetVector (Vector, NewLamda);
        Update (X, Vector);
        Display (it, NewLamda, X);
        Diff = fabs (NewLamda - OldLamda); it++;
    }
}

```

Teacher's Signature: _____

```
cout << endl;
```

```
cout << "largest Eigen Value Is :::" << NewLine << endl;
```

```
cout << "It's Eigen Vector Is :::" << ";
```

```
for (int i=0; i<3; i++) {
```

```
    if (i==2) { cout << Vector[i][0] << "]" << endl; }
```

```
    else { cout << Vector[i][0] << ", "; }
```

```
}
```

```
cout << endl;
```

```
return 0;
```

```
}
```

Teacher's Signature: _____

```

1 // Code To Implement Power Method On Matrix Using C++.....
2 // WAP To Find Largest Eigen value and corresponding Eigen vector using Power Metho
3 #include<iostream>
4 #include<cmath>
5 using namespace std;
6
7 double GetMax(double Temp[3][1]){
8     double Max = 0.0;
9     for (int i=0;i<3;i++){
10         if (Temp[i][0] > Max){ Max = Temp[i][0]; }
11     }
12     return Max;
13 }
14
15 void Multiply(double A[3][3] , double X[3][1] , double Result[3][1]){
16     for(int i=0;i<3;i++){ Result[i][0]=0; }
17     for(int i=0;i<3;i++){
18         for(int j=0;j<3;j++){
19             Result[i][0] += ((A[i][j])*(X[j][0]));
20         }
21     }
22 }
23
24 void SetVector(double Vector[3][1],double Lamda){
25     for(int i=0;i<3;i++){
26         Vector[i][0] = (Vector[i][0]/Lamda);
27     }
28 }
29
30 void Update(double X[3][1],double Result[3][1]){
31     for(int i=0;i<3;i++){
32         X[i][0]=Result[i][0];
33     }
34 }
35
36 void Display(int itr,double NewLamda,double X[3][1]){
37     cout<<"itr-<<itr<<"\t\t"<<NewLamda<<"\t\t [ ";
38     for(int i=0;i<3;i++){
39         if(i==2){ cout<<X[i][0]<<" ]"<<endl; }
40         else{ cout<<X[i][0]<<" , "; }
41     }
42 }
43
44 int main(){
45     cout<<endl;
46     double A[3][3]={2,-1,0},{-1,2,-1},{0,-1,2}};
47     double X[3][1]={1},{0},{0}};
48     double Vector[3][1]={0};
49     double OldLamda=1.0 , NewLamda=0.0;
50     double e=0.001,Diff=fabs(NewLamda-OldLamda);
51     int itr=1;
52     cout<<"itr"<<"\t\tEigen Value"<<"\t\tEigen Vector"<<endl;
53     cout<<endl;
54
55     while(Diff >= e){
56         Multiply(A,X,Vector);
57         OldLamda=NewLamda;
58         NewLamda=GetMax(Vector);
59         SetVector(Vector,NewLamda);
60         Update(X,Vector);
61         Display(itr,NewLamda,X);
62         Diff=fabs(NewLamda-OldLamda);
63         itr++;
64     }
65
66     cout<<endl;
67
68     cout<<"Largest Eigen Value Is ::: "<<NewLamda<<endl;
69     cout<<"It's Eigen Vector Will Be ::: [ ";
70     for(int i=0;i<3;i++){
71         if(i==2){ cout<<Vector[i][0]<<" ]"<<endl; }
72         else{ cout<<Vector[i][0]<<" , "; }
73     }
74
75     cout<<endl;
76
77     return 0;
78 }

```


OUTPUT :-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH TERMINAL OUTPUT  COMMENTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes\" ;
}

Itr      Eigen Value      Eigen Vector
itr-1    2      [ 1 , -0.5 , 0 ]
itr-2    2.5    [ 1 , -0.8 , 0.2 ]
itr-3    2.8    [ 1 , -1 , 0.428571 ]
itr-4    3      [ 1 , -1.14286 , 0.619048 ]
itr-5    3.14286  [ 1 , -1.24242 , 0.757576 ]
itr-6    3.24242  [ 1 , -1.30841 , 0.850467 ]
itr-7    3.30841  [ 1 , -1.35028 , 0.909605 ]
itr-8    3.35028  [ 1 , -1.37605 , 0.946037 ]
itr-9    3.37605  [ 1 , -1.39161 , 0.968032 ]
itr-10   3.39161  [ 1 , -1.40088 , 0.981149 ]
itr-11   3.40088  [ 1 , -1.40637 , 0.988914 ]
itr-12   3.40637  [ 1 , -1.40961 , 0.993491 ]
itr-13   3.40961  [ 1 , -1.41151 , 0.996182 ]
itr-14   3.41151  [ 1 , -1.41263 , 0.997762 ]
itr-15   3.41263  [ 1 , -1.41329 , 0.998688 ]
itr-16   3.41329  [ 1 , -1.41367 , 0.999231 ]

Largest Eigen Value Is ::: 3.41329
It's Eigen Vector Will Be ::: [ 1 , -1.41367 , 0.999231 ]

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-Labwork_Codes> |
```

PROGRAM - 11.

Q → WAP To Implement Differential Eqⁿ using Runge-Kutta Method?

```
→ #include <iostream>
#include <iomanip>
using namespace std;
```

```
double dydx (double x, double y) {
    return (x-y)/2; // dy/dx = (x-y)/2;
}
```

```
double Rungekuth (double x0, double y0, double x, double h) {
    int n = (int) ((x-x0)/h);
    double k1, k2, k3, k4, k;
    double y = y0;
    for (int i=0; i<n; i++) {
        k1 = h * dydx (x0, y);
        k2 = h * dydx (x0 + (0.5*h), y + (0.5*k1));
        k3 = h * dydx (x0 + (0.5*h), y + (0.5*k2));
        k4 = h * dydx (x0 + h, y + k3);
        k = (1/6.0) * (k1 + (2*k2) + (2*k3) + k4);
        y = y + k;
        x0 = x0 + h;
        cout << i << " " << x0 << " " << k1 << " " << k2
            << " " << k3 << " " << k4 << " " << y << endl;
    }
    return y;
}
```

Teacher's Signature: _____

```
int main () {
```

```
    cout << endl;
```

```
    double x0 = 0, y = 1, x = 2, h = 0.2;
```

```
    cout << "It is x k1 k2 k3 k4 y" << endl;
```

```
    double ans = RungeKutta (x0, y, x, h);
```

```
    cout << endl;
```

```
    cout << "The Value of y at x = " << x << " is :: " << ans << endl;
```

```
    cout << endl;
```

```
    return 0;
```

```
}
```

Teacher's Signature: _____


```

1 //Code To Implement Runge Kutta Method Of 4th Order Using C++.....
2 #include<iostream>
3 #include<iomanip>
4 using namespace std;
5
6 double dydx(double x, double y){ return((x - y)/2); } // "dy/dx = (x - y)/2".....
7
8 double RungeKutta(double x0, double y0, double x, double h){
9     int n = (int)((x - x0) / h);
10    double k1, k2, k3, k4, k5;
11    double y = y0;
12
13    for (int i=1; i<=n; i++){
14        k1 = h*dydx(x0, y);
15        k2 = h*dydx(x0 + 0.5*h, y + 0.5*k1);
16        k3 = h*dydx(x0 + 0.5*h, y + 0.5*k2);
17        k4 = h*dydx(x0 + h, y + k3);
18        y = y + (1.0/6.0)*(k1 + (2*k2) + (2*k3) + k4);
19        x0 = x0 + h;
20        cout<<i<<"\t"<<setw(8)<<x0<<"\t"<<setw(8)<<k1<<"\t"<<setw(8)<<k2<<"\t"<<setw(8)<<k3<<"\t"<<setw(8)<<k4<<"\t"<<setw(8)<<y<<endl;
21    }
22    return y;
23 }
24
25 int main(){
26     cout<<endl;
27     double x0 = 0, y = 1, x = 2, h = 0.2;
28     cout<<"Iter\t"<<setw(8)<<"x\t"<<setw(8)<<"k1\t"<<setw(8)<<"k2\t"<<setw(8)<<"k3\t"<<setw(8)<<"k4\t" <<setw(8)<<"y"<<endl;
29     double ans=RungeKutta(x0, y, x, h);
30     cout<<endl;
31     cout << "The value of y at x = "<<x<<" IS :: "<< ans<<endl;
32     cout<<endl;
33     return 0;
34 }
35

```

OUTPUT :-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SEARCH TERMINAL OUTPUT  COMMENTS

PS E:\Yash Pandey\YashPandey_Codes(DSA)> cd "e:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes\" ;
RungeKutta_Method }

Itr      x          k1          k2          k3          k4          y
1         0.2        -0.1        -0.085      -0.08575    -0.071425   0.914512
2         0.4        -0.0714512  -0.0578787 -0.0585573  -0.0455955  0.856193
3         0.6        -0.0456193  -0.0333383 -0.0339524  -0.022224   0.822455
4         0.8        -0.0222455  -0.0111333 -0.0116889  -0.00107664 0.810961
5         1         -0.00109609 0.00895872 0.00845598 0.0180583   0.819593
6         1.2        0.0180407   0.0271387 0.0266838 0.0353723   0.846436
7         1.4        0.0353564   0.0435886 0.043177 0.0510387   0.889757
8         1.6        0.0510243   0.0584731 0.0581007 0.0652142   0.947988
9         1.8        0.0652012   0.0719412 0.0716042 0.0780408   1.01971

The value of y at x = 2 IS ::: 1.01971

PS E:\Yash Pandey\YashPandey_Codes(DSA)\C++_Codes\CM-LabWork_Codes>
```

